

COURIER MANAGEMENT SYSTEM (CMS)

TITLE-WISE PROJECT IMPLEMENTATION DOCUMENTATION

A Salesforce-based Courier and Logistics Management Application

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1. PROJECT OVERVIEW

The Courier Management System (CMS) is a Salesforce-based application designed to digitize and centralize courier and logistics operations. The implementation brings together customer management, shipment booking and tracking, delivery agent assignment, branch management, delivery updates, invoice information, validation, automation, reports and dashboards.

1.1 Six Custom Objects

Object	Purpose
Customer__c	Stores customer personal and contact information.
Shipment__c	Manages parcel details, source, destination, and tracking information.
Delivery_Agent__c	Maintains delivery agent profiles and branch details.
Branch__c	Represents courier branch offices.
Delivery_Update__c	Stores status updates for each shipment.
Invoice__c	Manages billing details for shipments.

2. PHASE 1 – SALESFORCE ACCOUNT AND ORG SETUP

The Salesforce Developer environment was established first, providing access to Setup, Object Manager, Lightning App configuration and the other Salesforce administration features used in the project.

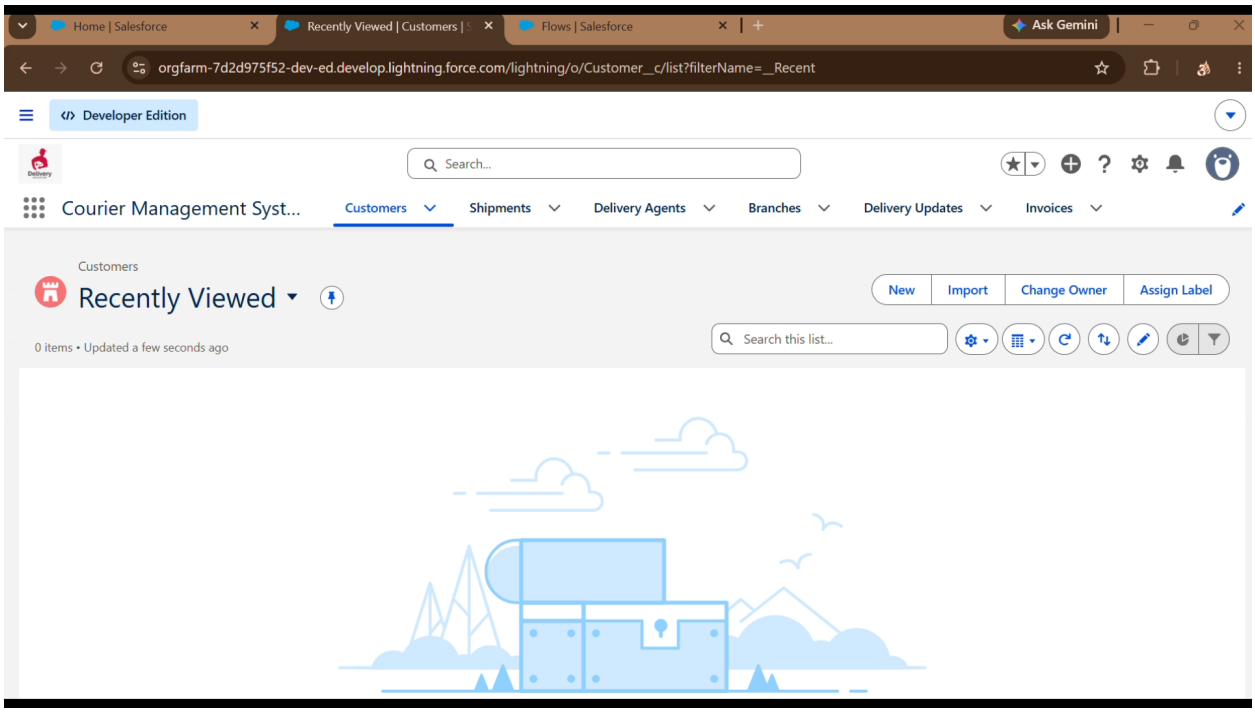


Figure 1 – Salesforce CMS application after the Salesforce org setup.

3. PHASE 2 – CUSTOM OBJECT CREATION

Six custom objects were created to represent the CMS entities. The Object Manager screenshot provides direct evidence of the Salesforce data-model implementation.

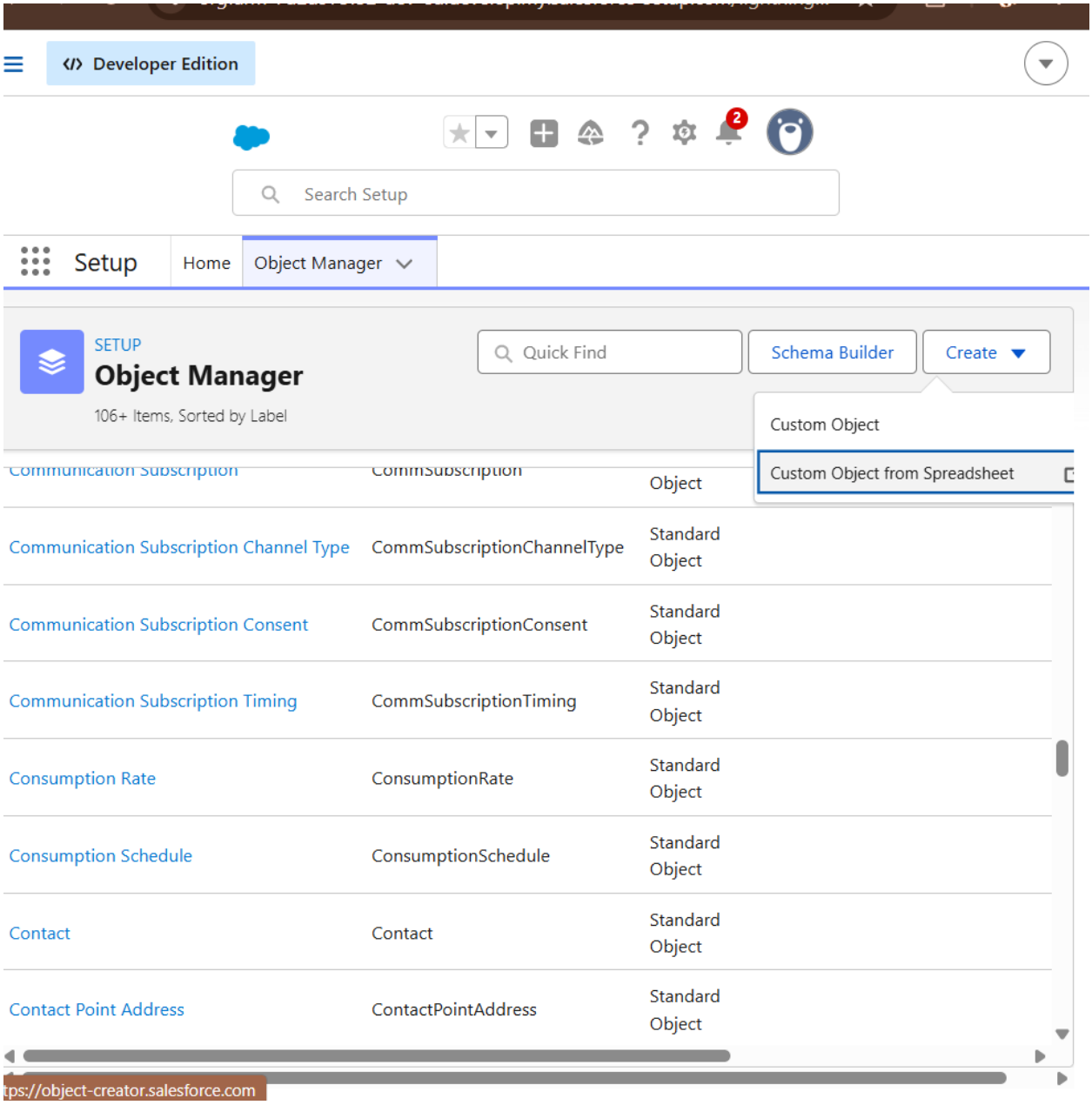


Figure 2 – Salesforce Object Manager showing the CMS custom-object implementation.

Objects Created

- **Customer__c** – Stores customer personal and contact information.
- **Shipment__c** – Manages parcel details, source, destination, and tracking information.
- **Delivery_Agent__c** – Maintains delivery agent profiles and branch details.
- **Branch__c** – Represents courier branch offices.
- **Delivery_Update__c** – Stores status updates for each shipment.

• **Invoice__c** – Manages billing details for shipments.

4. PHASE 3 – CUSTOM TABS

Custom tabs were created so the CMS objects could be accessed through the Salesforce application. The project navigation uses Customers, Shipments, Delivery Agents, Branches, Delivery Updates and Invoices.

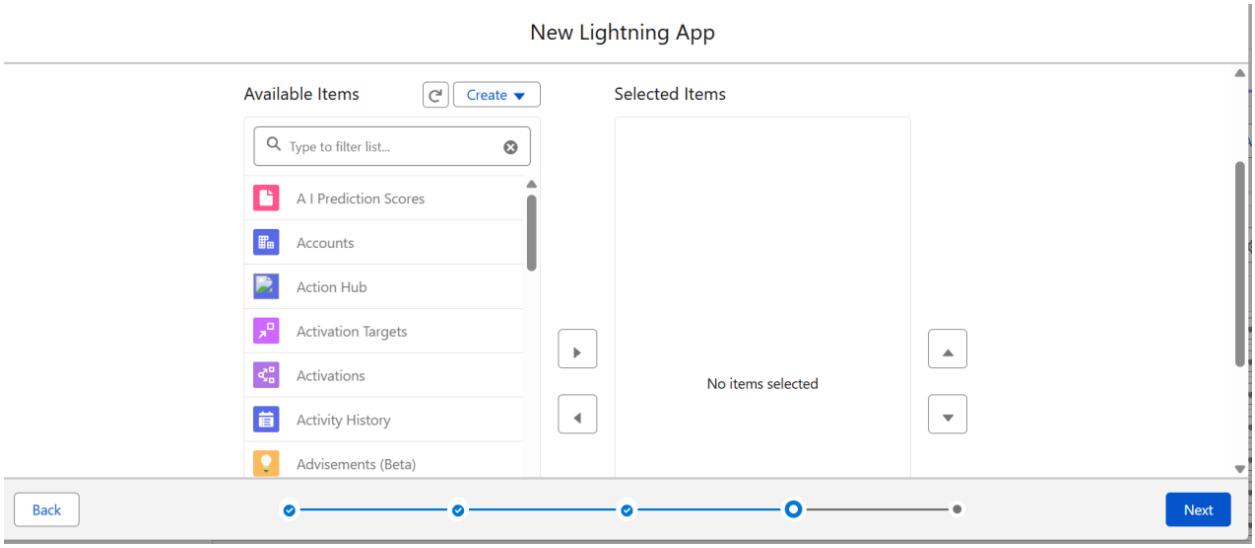


Figure 3 – Navigation Items configured for the Courier Management System.

5. PHASE 4 – FIELD CREATION AND RELATIONSHIPS

Field creation was a major implementation phase. Each object is presented under its own title, followed by its field table and the available Salesforce screenshot evidence.

5.1 Customer__c – Customer Fields

Field	API Name	Type	Required
Customer Name	Name	Text	Yes
Email	Email__c	Email	Yes
Phone	Phone__c	Phone	Yes
Address	Address__c	Text Area	Yes
Customer Type	Customer_Type__c	Picklist	Yes
City	City__c	Text	No
State	State__c	Text	No
Pincode	Pincode__c	Text	No
Is Active	Is_Active__c	Checkbox	No

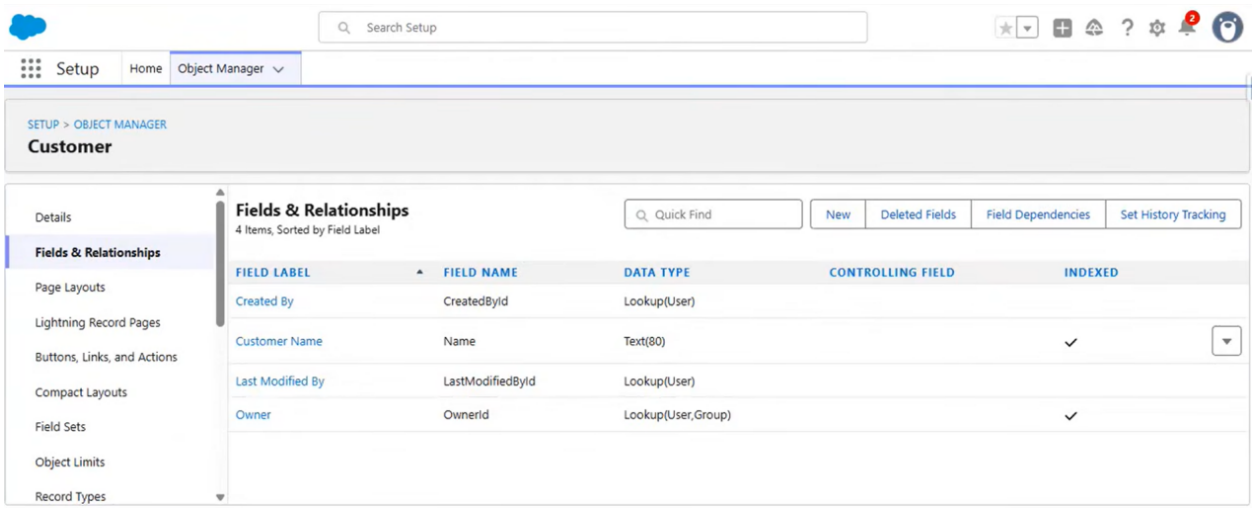


Figure 4 – Customer__c Fields & Relationships.

5.2 Shipment__c – Shipment Fields

Field	API Name	Type	Required
Tracking Number	Name	Auto Number	Yes
Customer	Customer__c	Lookup(Customer)	Yes
Source Address	Source_Address__c	Text Area	Yes
Destination Address	Destination_Address__c	Text Area	Yes
Weight	Weight__c	Number	Yes
Price	Price__c	Currency	Yes
Status	Status__c	Picklist	Yes

Field	API Name	Type	Required
Expected Date	Expected_Date__c	Date	Yes
Assigned Agent	Assigned_Agent__c	Lookup(Delivery Agent)	No
Shipment Type	Shipment_Type__c	Picklist	Yes

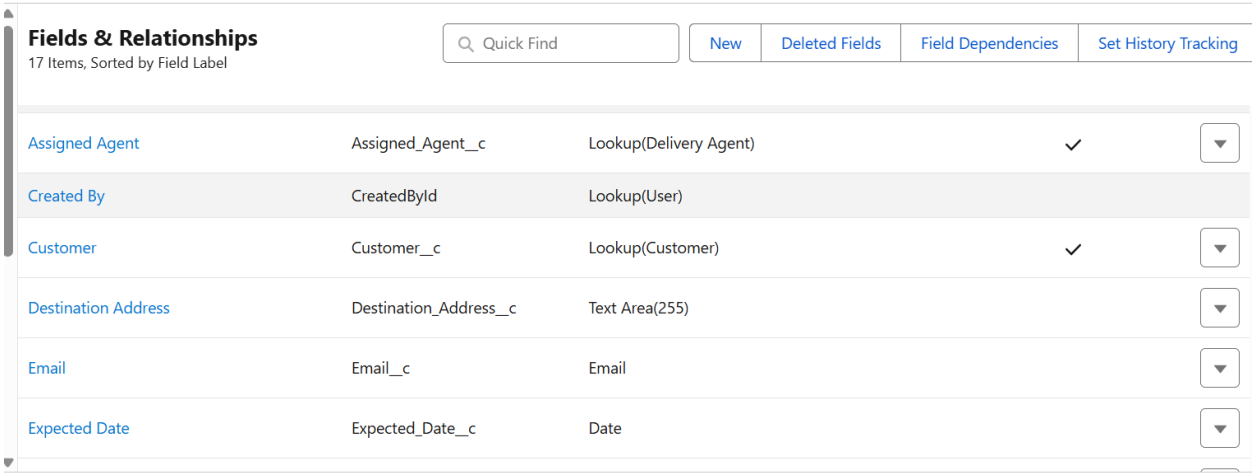


Figure 5 – Shipment__c Fields & Relationships.

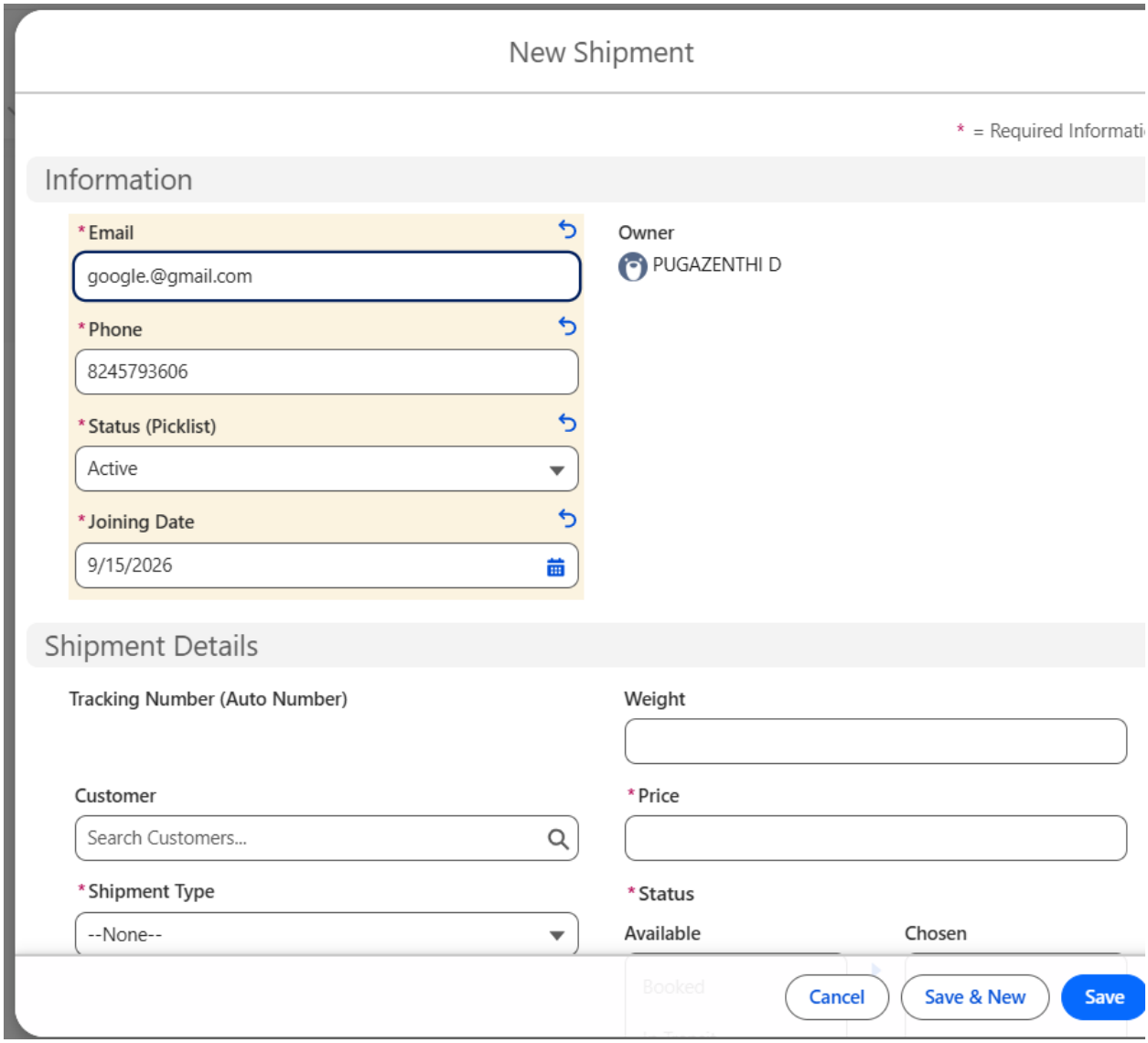


Figure 6 – Shipment record creation evidence.

5.3 Delivery_Agent__c – Delivery Agent Fields

Field	API Name	Type	Required
Agent Name	Name	Text	Yes
Email	Email__c	Email	Yes
Phone	Phone__c	Phone	Yes
Branch	Branch__c	Lookup(Branch)	Yes
Status	Status__c	Picklist	Yes
Joining Date	Joining_Date__c	Date	No

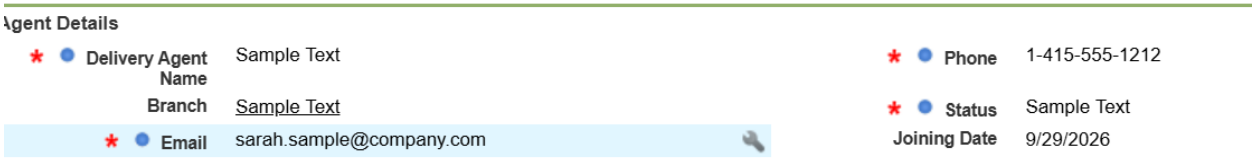


Figure 7 – Delivery Agent fields and record-layout evidence.

5.4 Branch__c – Branch Fields

Field	API Name	Type	Required
Branch Name	Name	Text	Yes
Branch Code	Branch_Code__c	Text	Yes
City	City__c	Text	Yes
State	State__c	Text	Yes
Manager Name	Manager_Name__c	Text	No

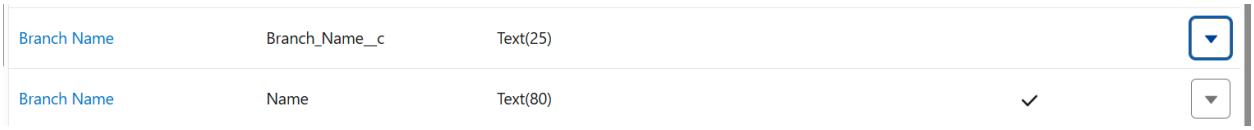


Figure 8 – Branch__c fields and field configuration.

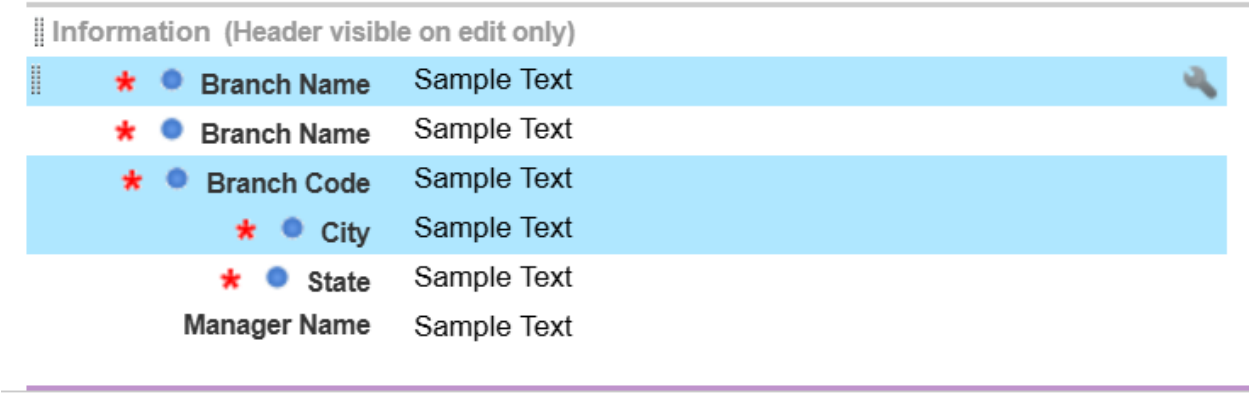


Figure 9 – Branch page-layout evidence.

5.5 Delivery_Update__c – Delivery Update Fields

Field	API Name	Type	Required
Update Number	Name	Auto Number	Yes
Shipment	Shipment__c	Master-Detail(Shipment)	Yes
Status	Status__c	Picklist	Yes
Update Date	Update_Date__c	DateTime	Yes
Remarks	Remarks__c	Text	No

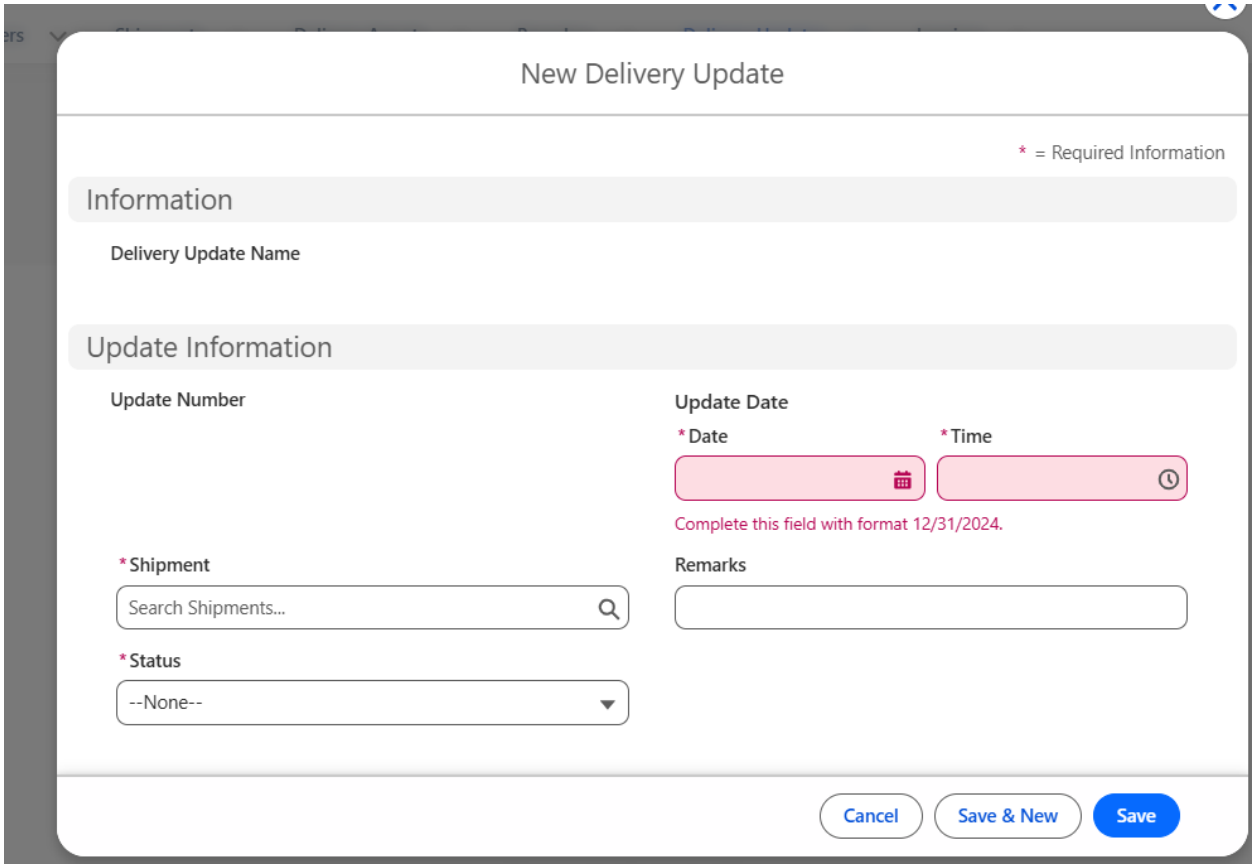


Figure 10 – Delivery Update record creation screen.

5.6 Invoice__c – Invoice Fields

Field	API Name	Type	Required
Invoice Number	Name	Auto Number	Yes
Shipment	Shipment__c	Lookup(Shipment)	Yes
Amount	Amount__c	Currency	Yes
Invoice Date	Invoice_Date__c	Date	Yes
Payment Status	Payment_Status__c	Picklist	Yes

No dedicated Invoice Fields & Relationships screenshot was available in the supplied screenshot set; the field configuration is therefore documented by the table without inventing an image.

5.7 Object Relationships

- Customer → Shipment: Lookup relationship.
- Shipment → Delivery Agent: Lookup relationship.
- Delivery Agent → Branch: Lookup relationship.
- Delivery Update → Shipment: Master-Detail relationship.
- Invoice → Shipment: Lookup relationship.

These relationships form the CMS data flow from customer and shipment records through delivery operations and billing.

6. PHASE 5 – VALIDATION RULES

Validation rules were created to prevent invalid customer, shipment, delivery-agent, branch and delivery-update data.

Rule Name	Object	Condition	Error Message
Email_Required	Customer__c	ISBLANK(Email__c)	Email is required.
Validate_Phone_Number	Customer__c	LEN(Phone__c) <> 10	Phone number must be 10 digits.
Validate_Pincode	Customer__c	LEN(Pincode__c) <> 6	Pincode must be 6 digits.
Validate_Weight	Shipment__c	Weight__c <= 0	Weight must be greater than zero.
Validate_Price	Shipment__c	Price__c <= 0	Price must be greater than zero.
Validate_Expected_Date	Shipment__c	Expected_Date__c < TODAY()	Expected delivery date cannot be in the past.
Validate_Agent_Email	Delivery_Agent__c	NOT(CONTAINS(Email__c, "@"))	Enter a valid email address.
Branch_Code_Required	Branch__c	ISBLANK(Branch_Code__c)	Branch Code is mandatory.
Remarks_Required_For_Delay	Delivery_Update__c	AND(ISPICKVAL(Status__c, "Delayed"), ISBLANK(Remarks__c))	Remarks are required when status is Delayed.

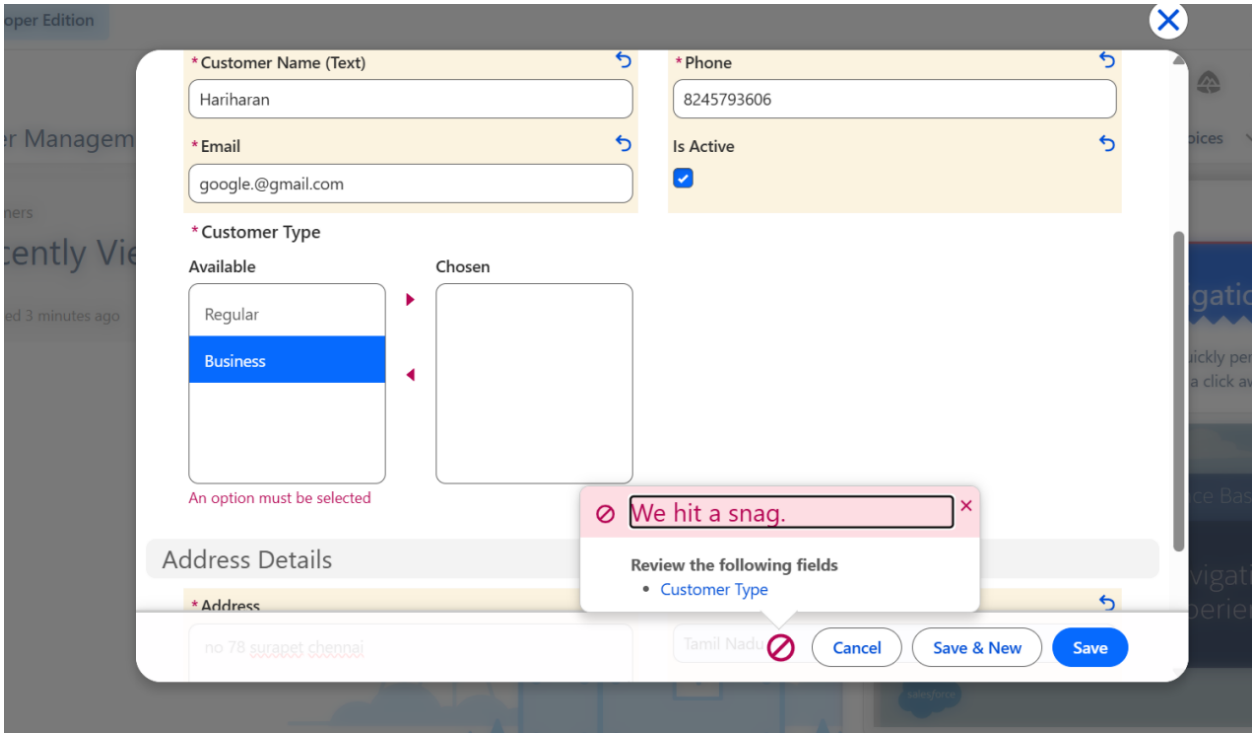


Figure 11 – Customer validation / required-field error evidence.

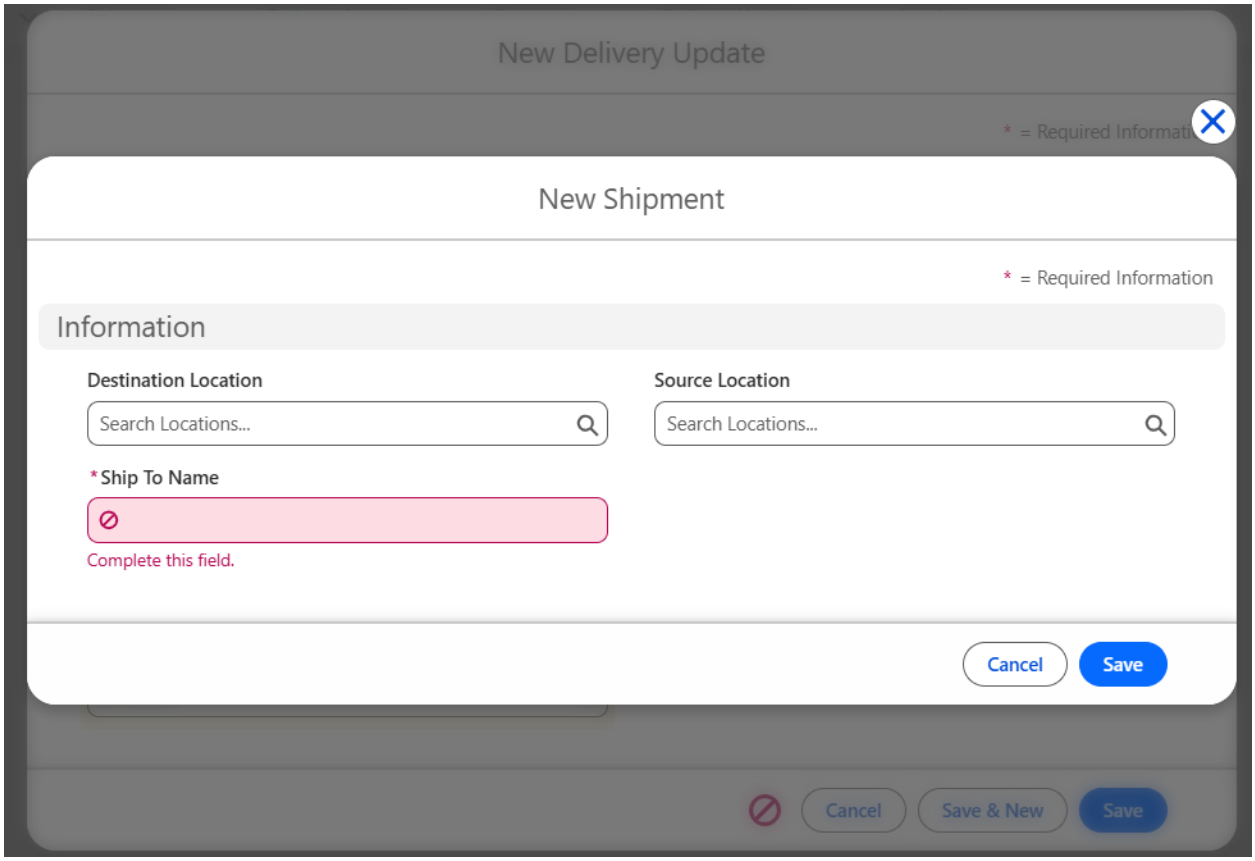


Figure 12 – Required-field validation evidence during shipment creation.

7. PHASE 6 – PAGE LAYOUTS

Page layouts were organized so that required information is presented in functional sections and related information is accessible in the Lightning record interface.

Object	Configured Layout Sections
Customer__c	Customer Information; Address Details
Shipment__c	Shipment Details; Address Information; Delivery Information
Delivery_Agent__c	Agent Details; Related Shipments
Branch__c	Branch Information
Delivery_Update__c	Update Information
Invoice__c	Invoice Details

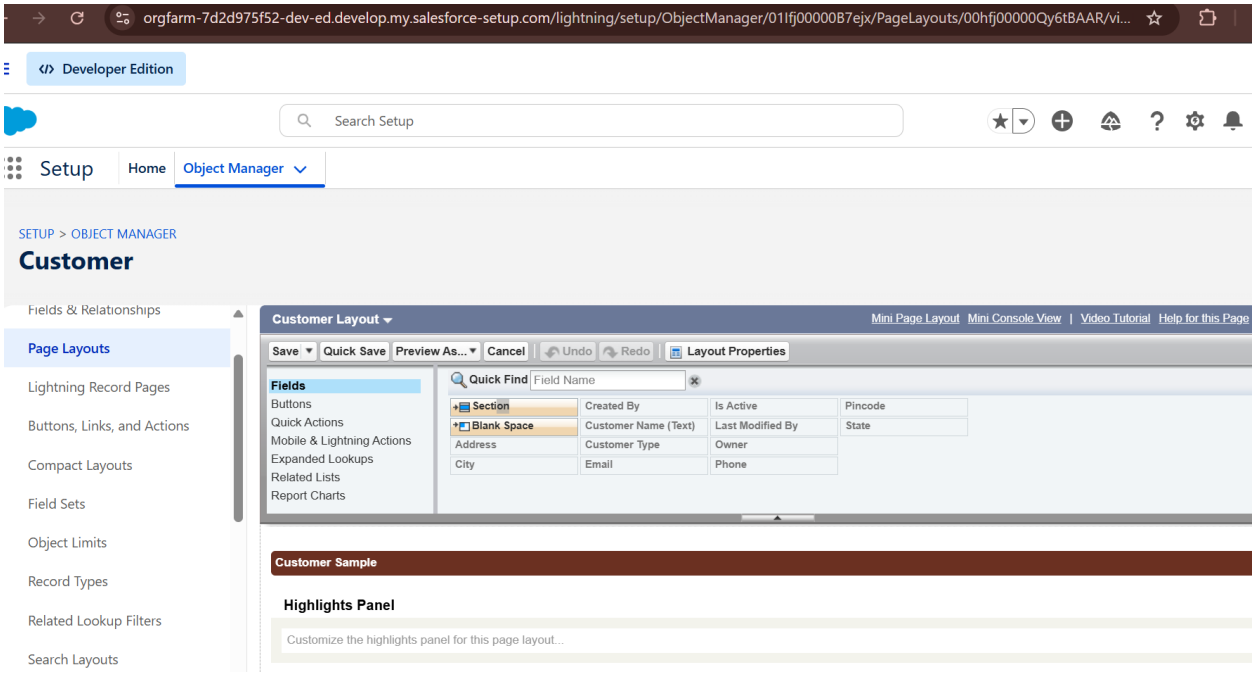


Figure 13 – Customer Layout configuration.

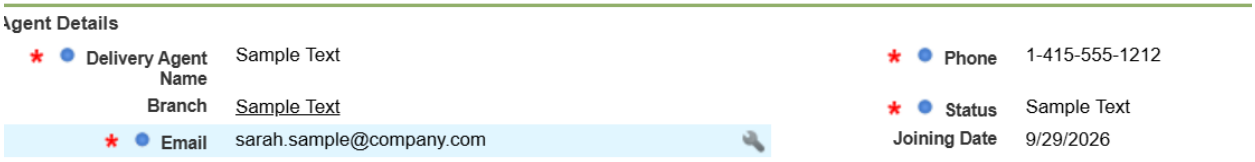


Figure 14 – Delivery Agent layout evidence.

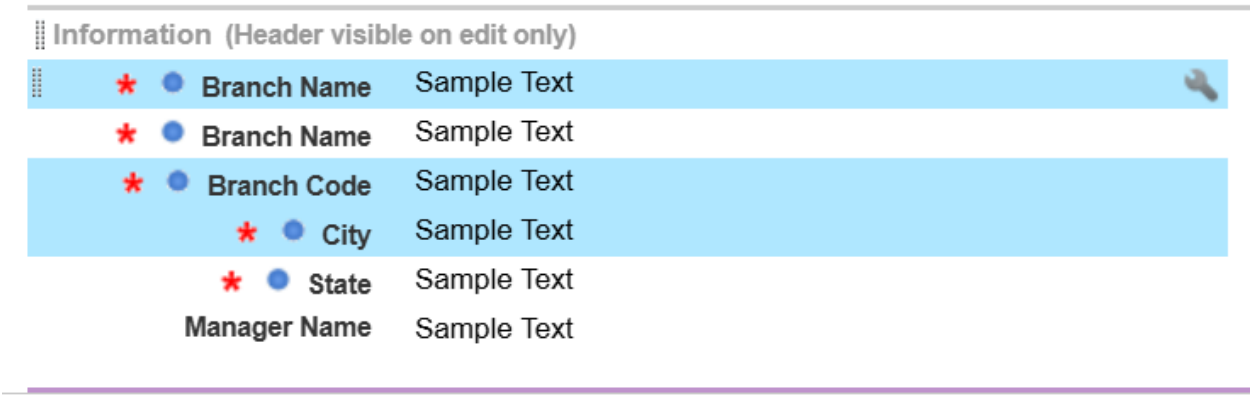


Figure 15 – Branch layout evidence.

8. PHASE 7 – FLOW CREATION AND AUTOMATION

A record-triggered Flow named Auto_Update_Shipment_Status was implemented for Delivery_Update__c. When a Delivery Update is created, the related Shipment status is updated using the Delivery Update status.

8.1 Flow Creation – Start Configuration

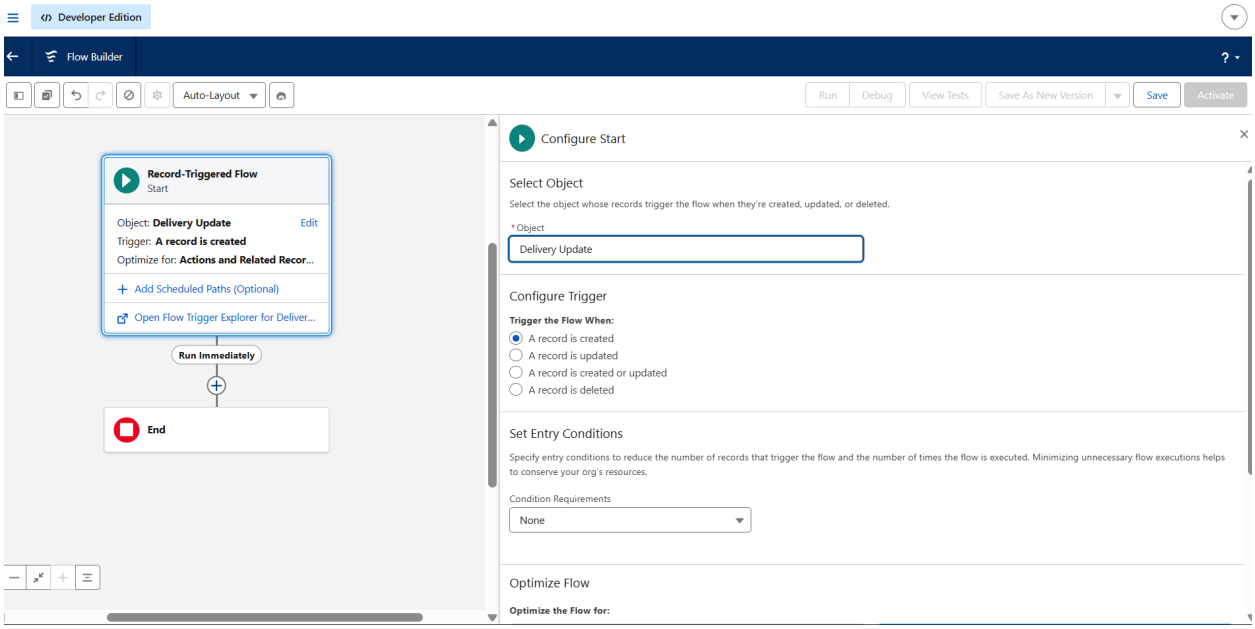


Figure 16 – Record-triggered Flow start configuration for Delivery Update.

8.2 Final Flow Canvas

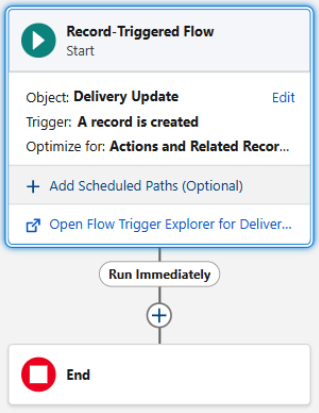


Figure 17 – Final Auto_Update_Shipment_Status Flow canvas.

Flow sequence: Delivery_Update__c record created → update related Shipment status → End.

8.3 Development Evidence

An intermediate Flow configuration screenshot was captured during development. It is retained as development evidence and is not treated as the final Flow design.

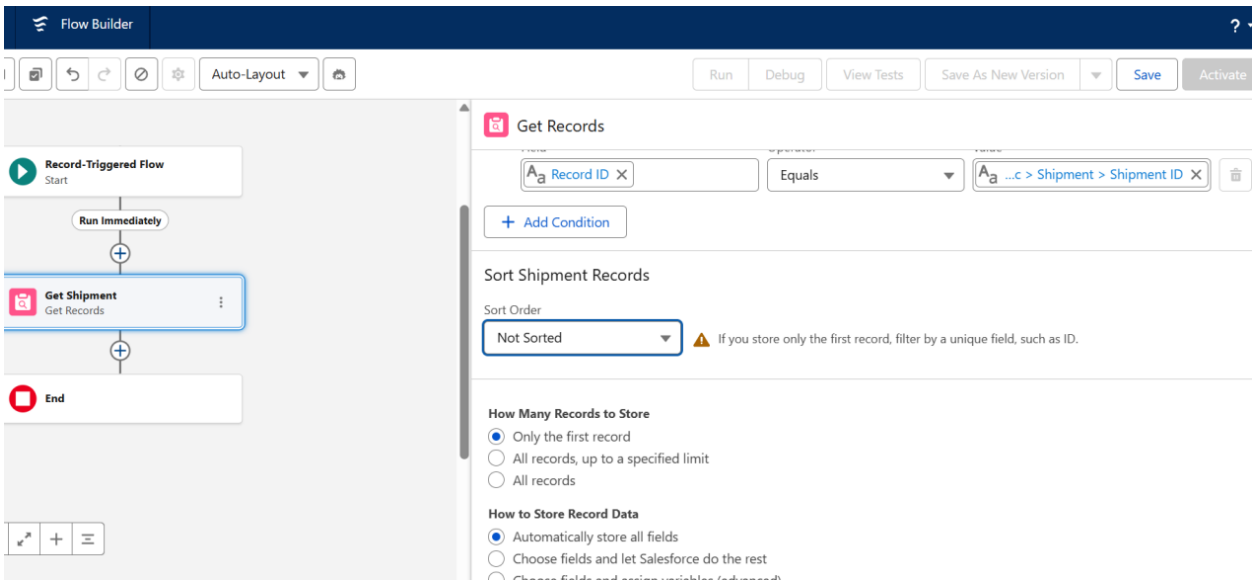


Figure 18 – Intermediate Flow configuration captured during development.

9. PHASE 8 – LIGHTNING APP CREATION

The Courier Management System Lightning App was created and configured with standard navigation, supported desktop and phone form factors, CMS navigation items and selected user profiles.

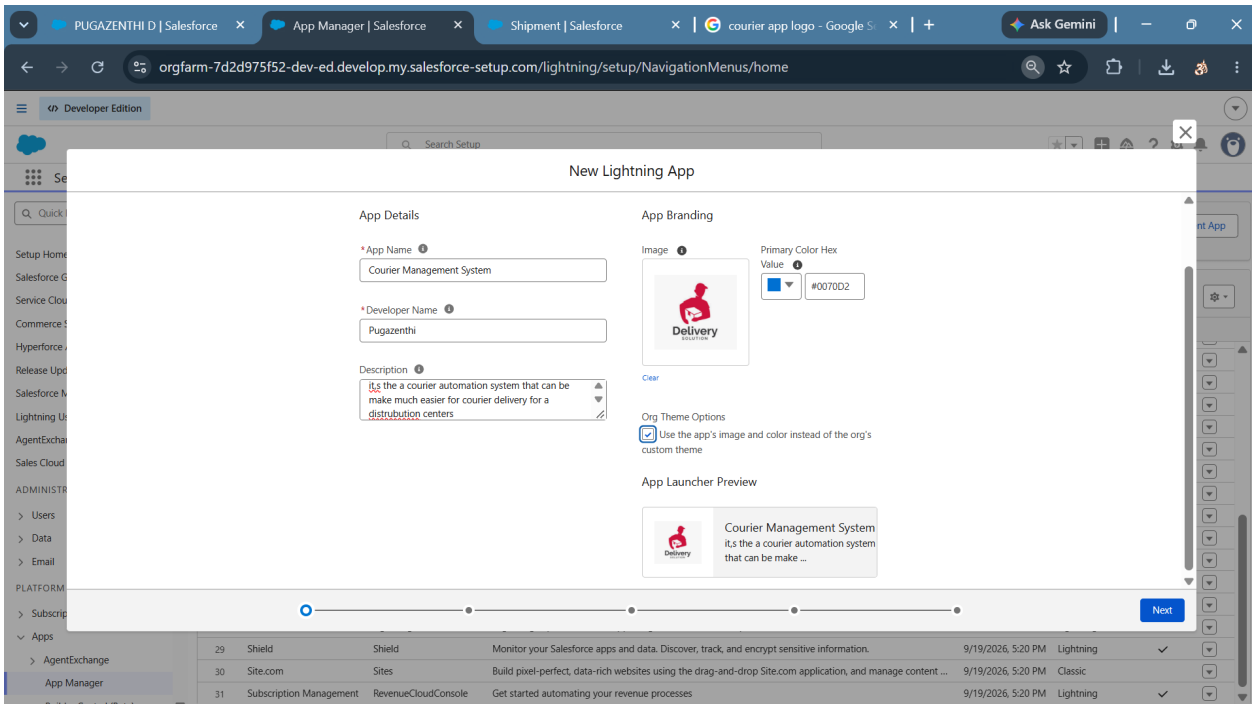


Figure 19 – New Lightning App details and branding.

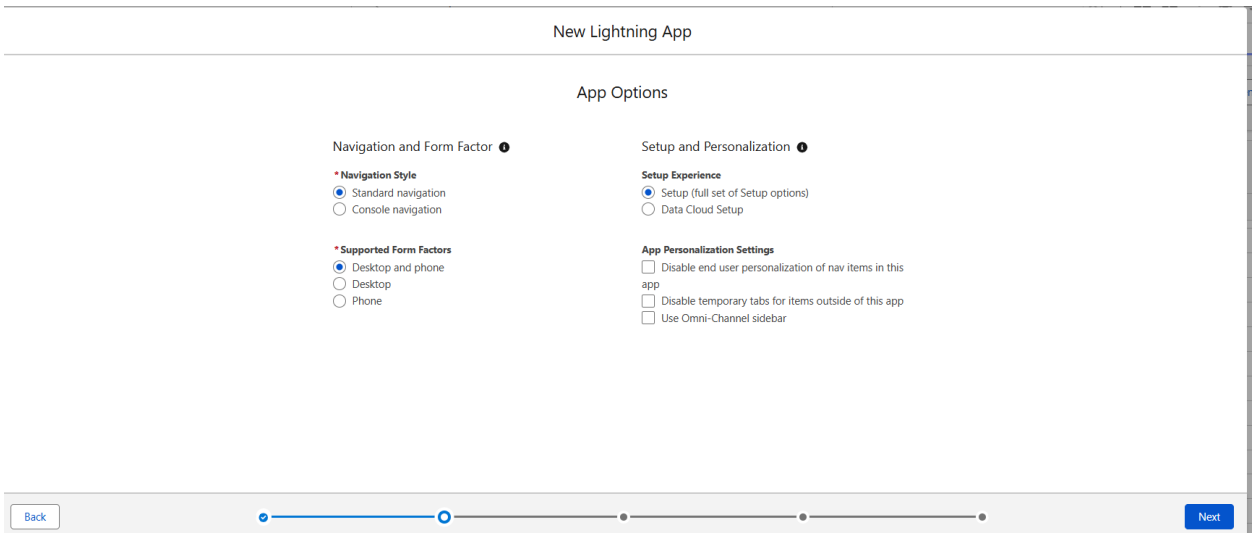


Figure 20 – Lightning App options.

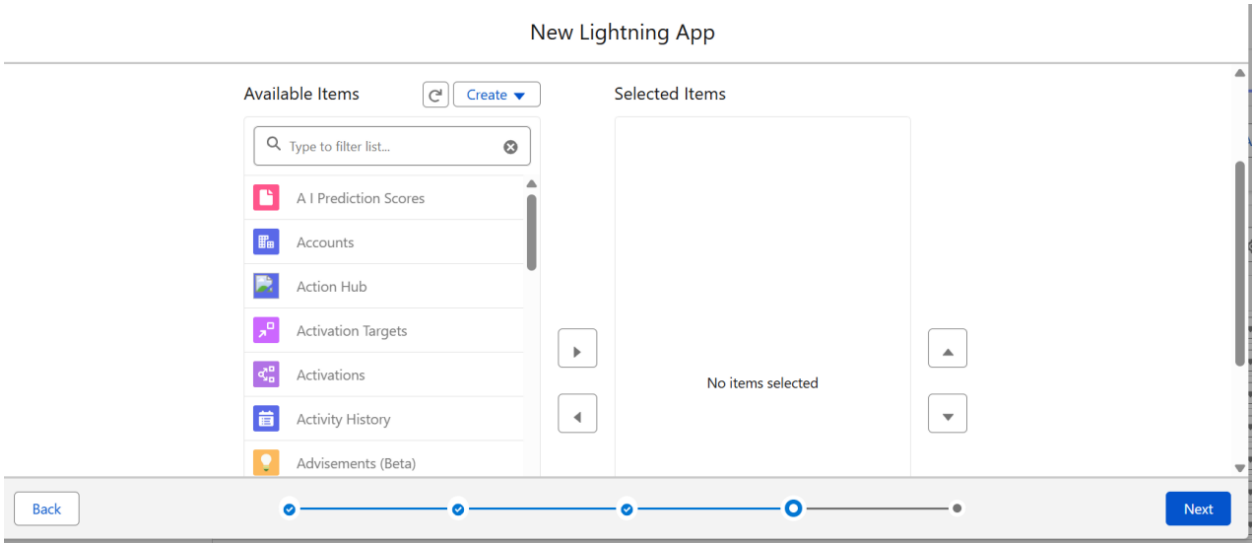


Figure 21 – Navigation Items configuration.

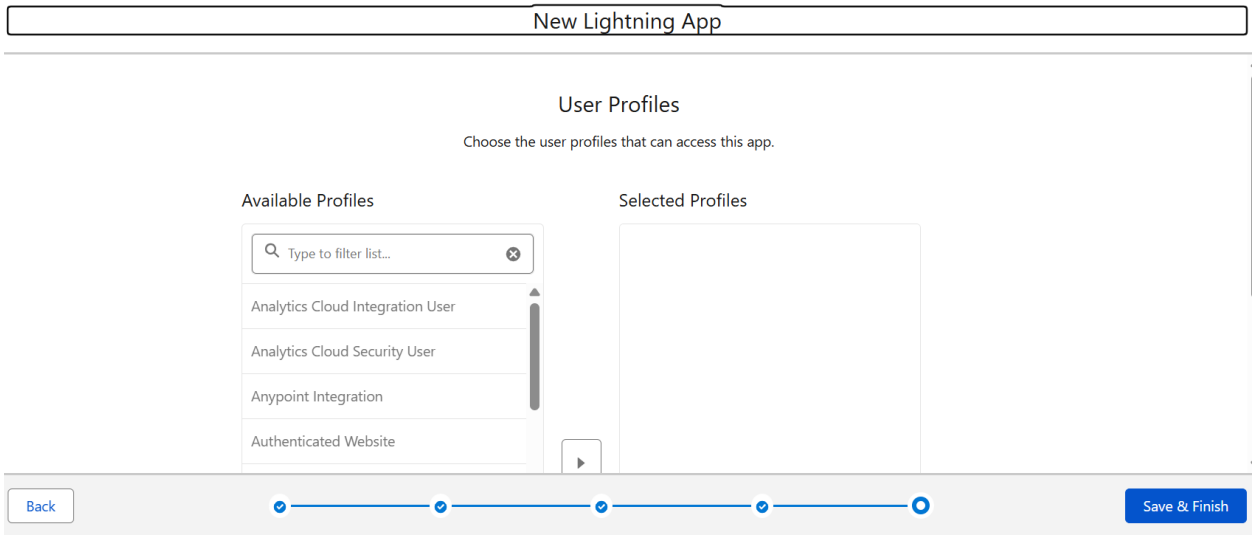


Figure 22 – User Profiles selected for the application.

10. PHASE 9 – PROFILES

The project created four profiles: Courier Admin, Branch Manager, Delivery Agent and Customer Support. Profiles control object permissions, tab access and other user-level settings.

- Courier Admin
- Branch Manager
- Delivery Agent
- Customer Support

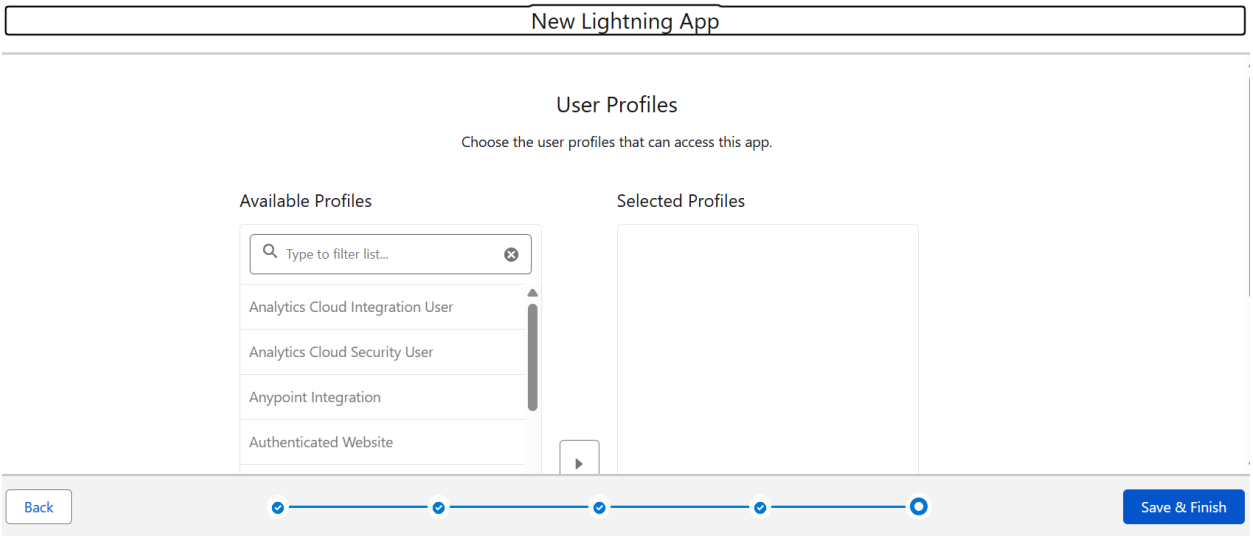


Figure 23 – Profiles selected for the Courier Management System app.

11. PHASE 10 – ROLES

The role hierarchy was configured as follows:

CEO → Courier Admin → Branch Manager → Delivery Agent / Customer Support

Courier Admin is the highest CMS administration role; Branch Manager manages branch-level operations; Delivery Agent handles assigned shipment operations; Customer Support provides customer and shipment tracking support.

12. PHASE 11 – USERS

Users were created and assigned the appropriate role, profile and license according to the task document. The four project roles are Courier Admin, Branch Manager, Delivery Agent and Customer Support.

Profiles control permissions while roles are used in the data-visibility hierarchy.

13. PHASE 12 / MILESTONE 15 – REPORTS AND DASHBOARDS

The source task document identifies the reporting and dashboard work as Milestone 15. The implementation includes report types, operational reports and the Courier Management Dashboard.

13.1 Report Types

Report Type	Primary Object	Purpose
Shipment Tracking Report	Shipment__c	Analyze shipment tracking and operational shipment data.
Customer Summary Report	Customer__c	Summarize customer information.
Agent Performance Report	Delivery_Agent__c	Analyze delivery-agent performance.

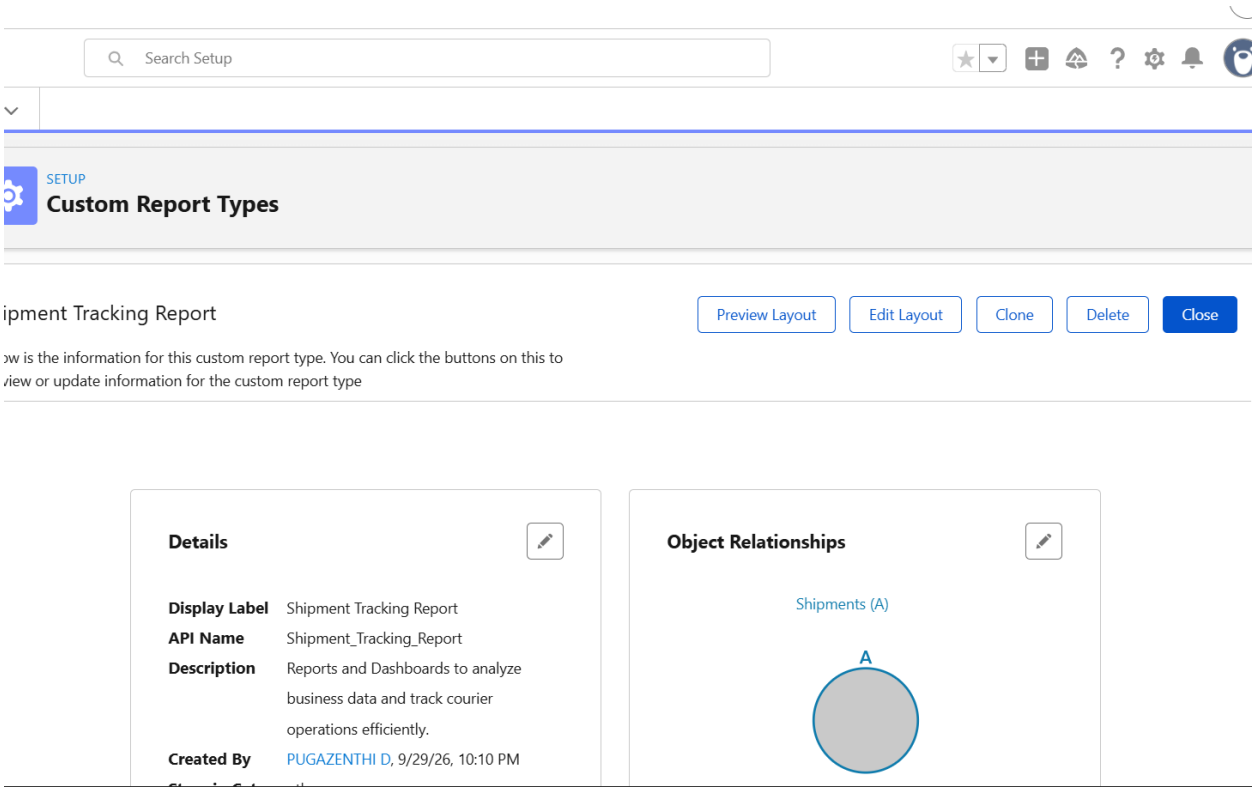


Figure 24 – Shipment Tracking Report custom report type details.

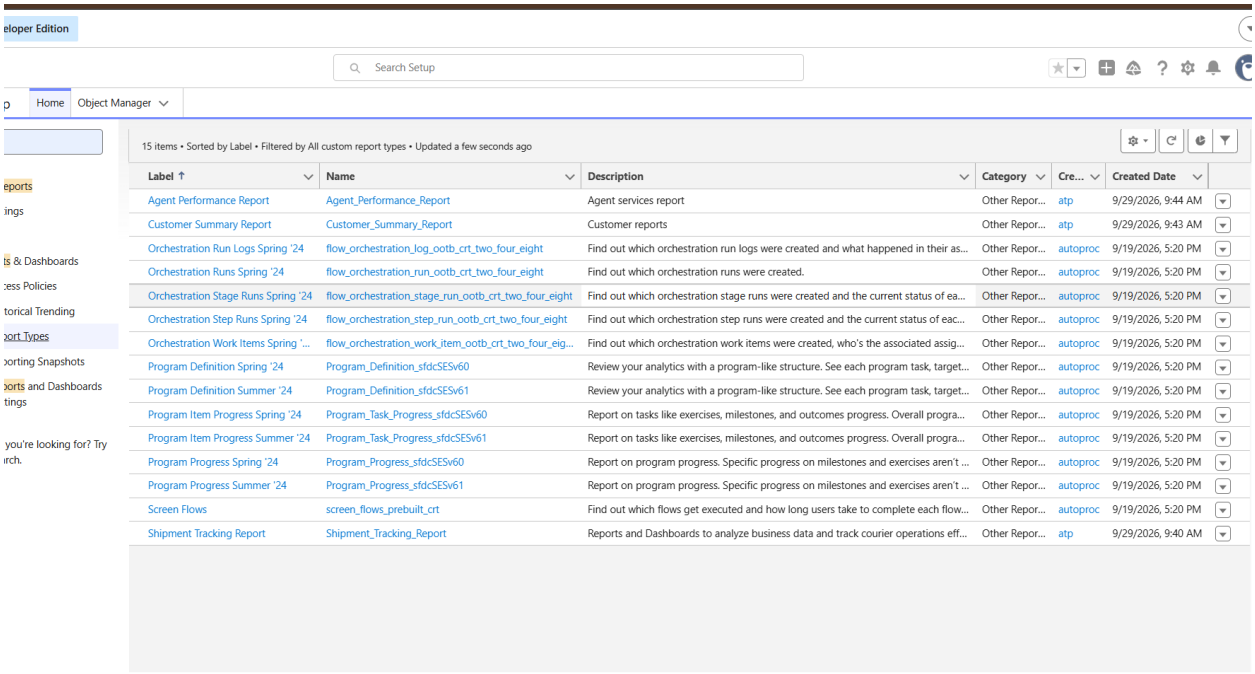


Figure 25 – Custom Report Types list showing CMS report types.

13.2 Shipment Status Summary / Shipment Tracking Report

The report uses tracking number, customer, status, expected date and assigned agent information and groups shipment records by status.

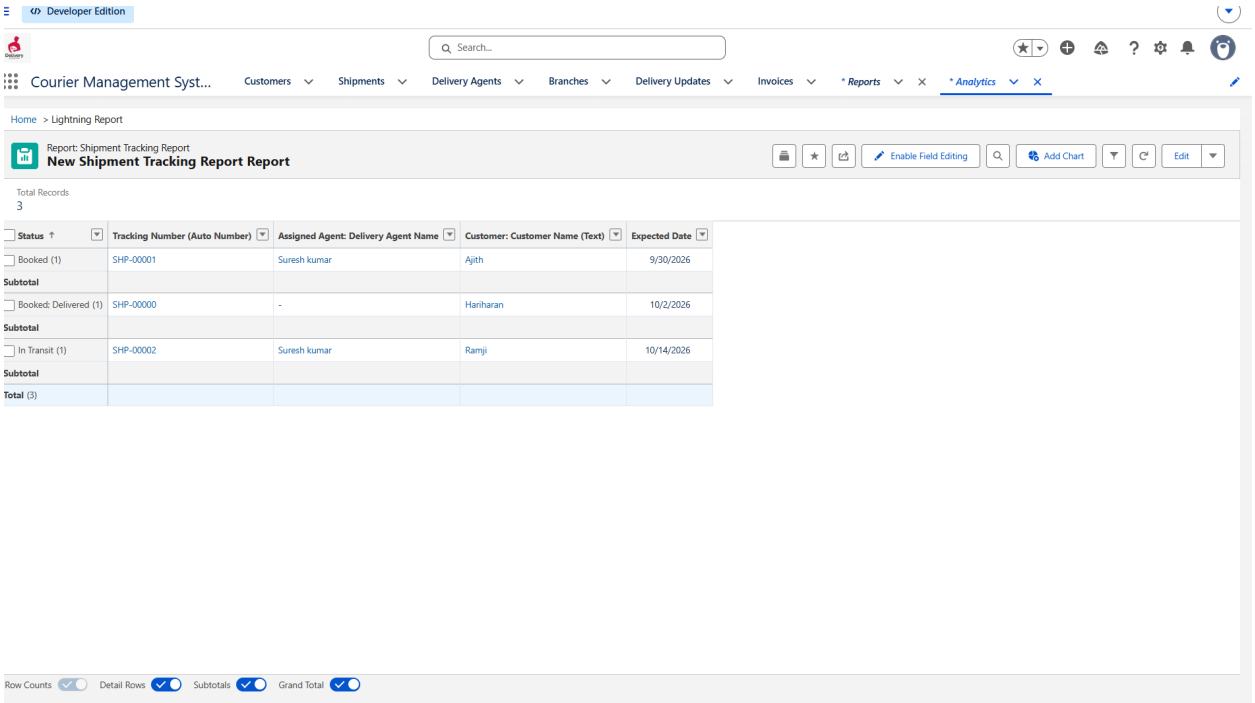


Figure 26 – Shipment Tracking Report output.

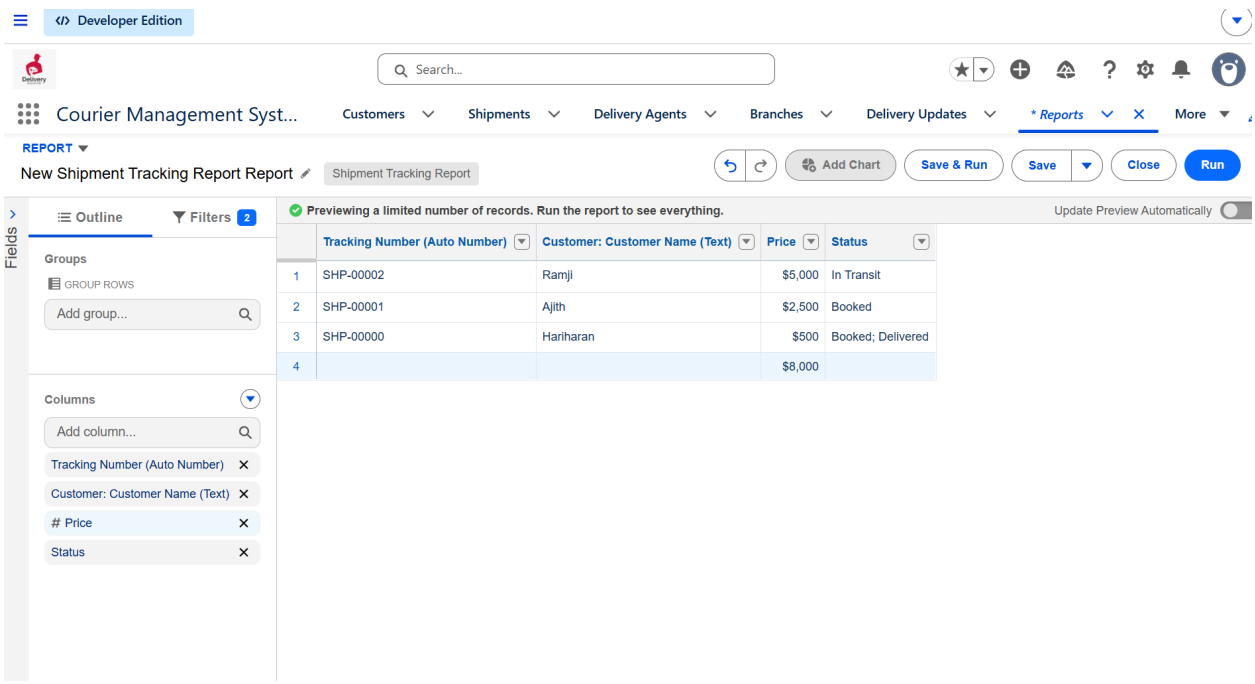


Figure 27 – Report builder showing fields and report configuration.

13.3 Revenue by Branch Report

The Revenue by Branch report was created as part of the reporting work. The report-list evidence below confirms the report exists in the CMS Reports folder.

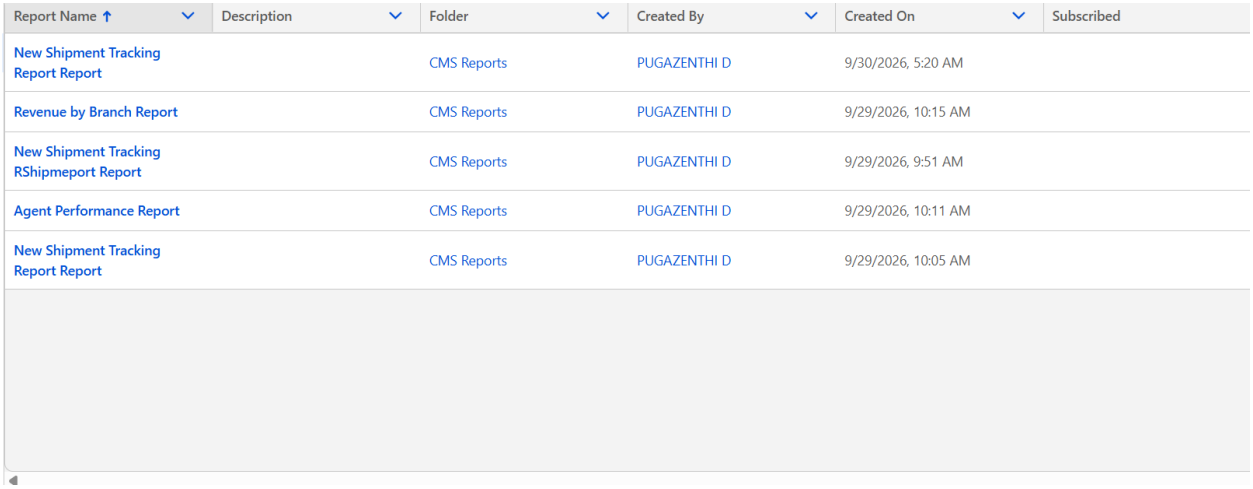


Figure 28 – Reports list including Revenue by Branch Report.

13.4 Dashboard Creation

The Courier Management Dashboard was configured with Shipment Status Overview (Donut Chart), Agent Performance (Bar Chart), and Revenue by Branch (Column Chart).

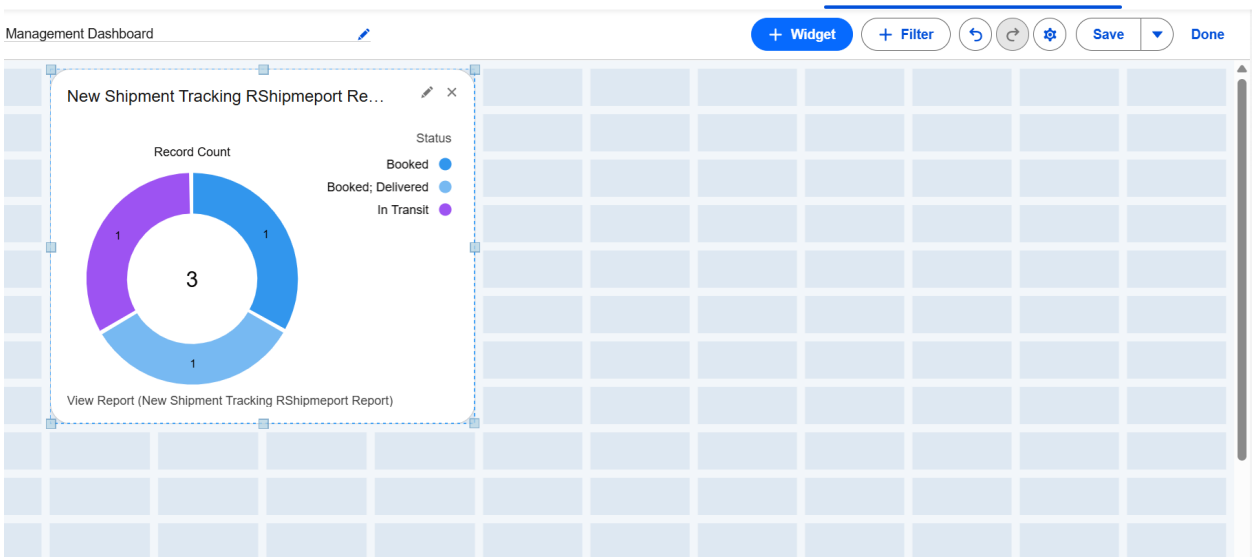


Figure 29 – Dashboard widget configuration screen.

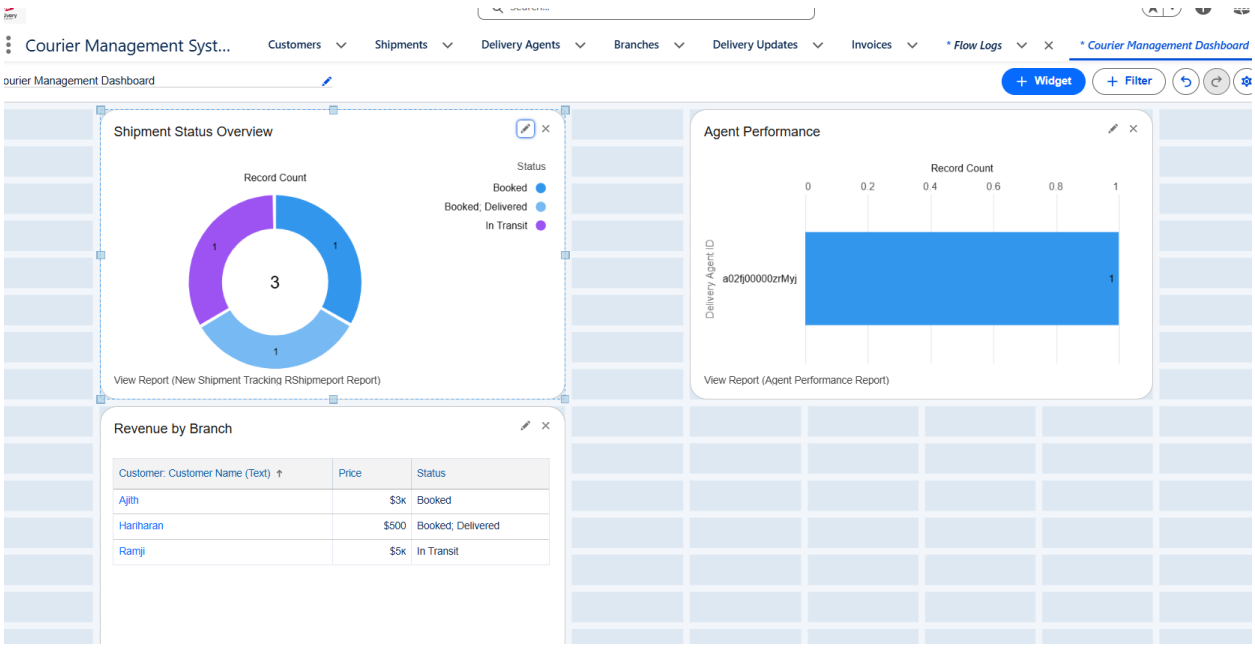


Figure 30 – Courier Management Dashboard with Shipment Status Overview, Agent Performance and Revenue by Branch.

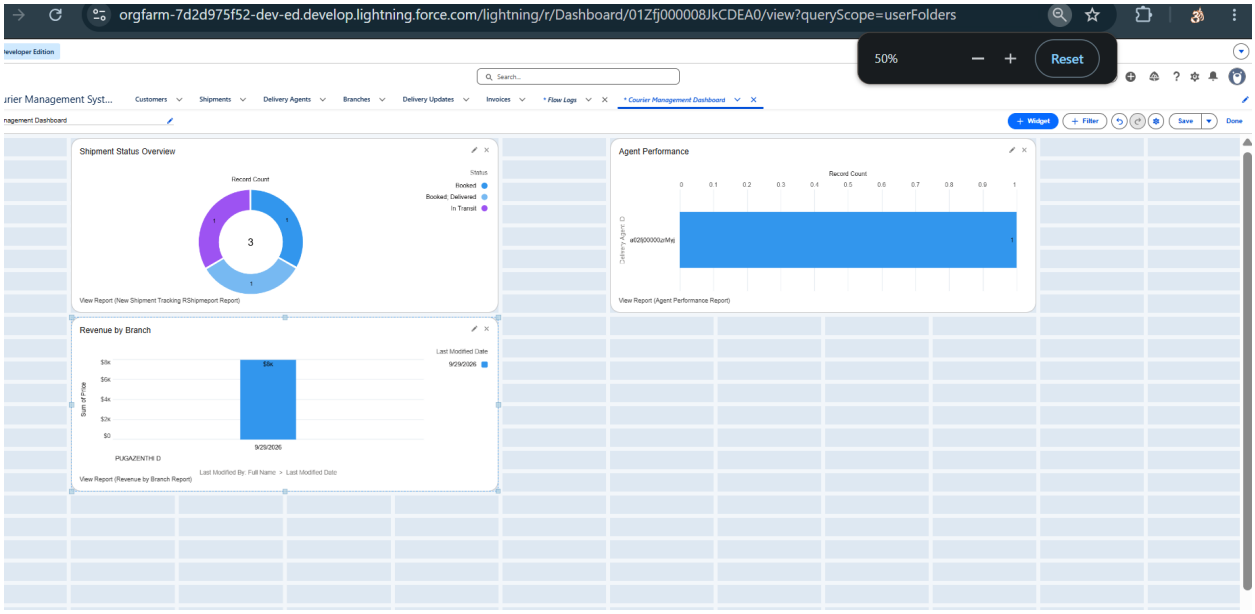


Figure 31 – Dashboard view showing the configured operational charts.

13.5 Supporting Report Configuration Evidence

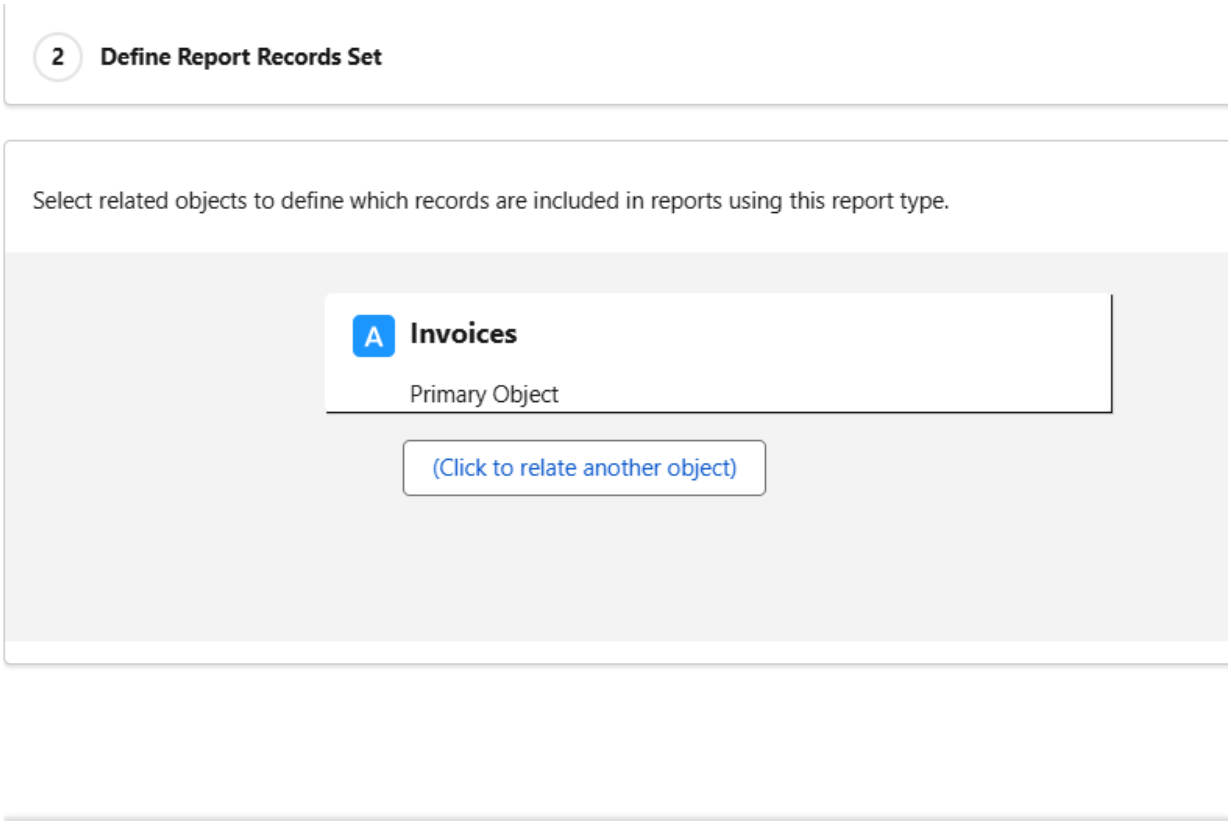


Figure 32 – Invoice-based report-type configuration evidence.

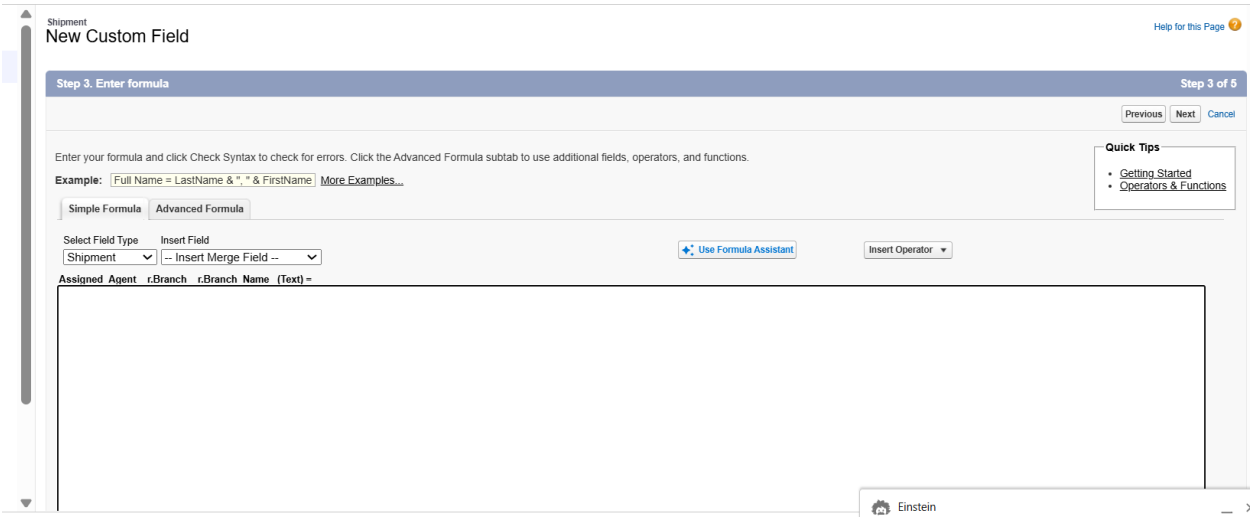


Figure 33 – Formula field configuration used during report development.

14. FUNCTIONAL AND PERFORMANCE TESTING

Test Scenario	Verification
Customer record creation	Create a valid customer record and verify that the record is saved correctly.
Customer validation	Verify email, phone number, and pincode validation rules.
Shipment validation	Verify weight, price, and expected delivery date validation rules.
Agent validation	Verify invalid email input is rejected.
Branch validation	Verify Branch Code cannot be blank.
Delivery Update validation	Verify remarks are required when status is Delayed.
Flow automation	Create a Delivery Update and verify that the related Shipment status is updated automatically.
Invoice generation	Create an invoice linked to a shipment and verify invoice details.
Reports	Run reports and verify displayed fields and grouping.

The screenshot evidence demonstrates record creation, validation errors, Flow configuration and reporting/dashboard visualization.

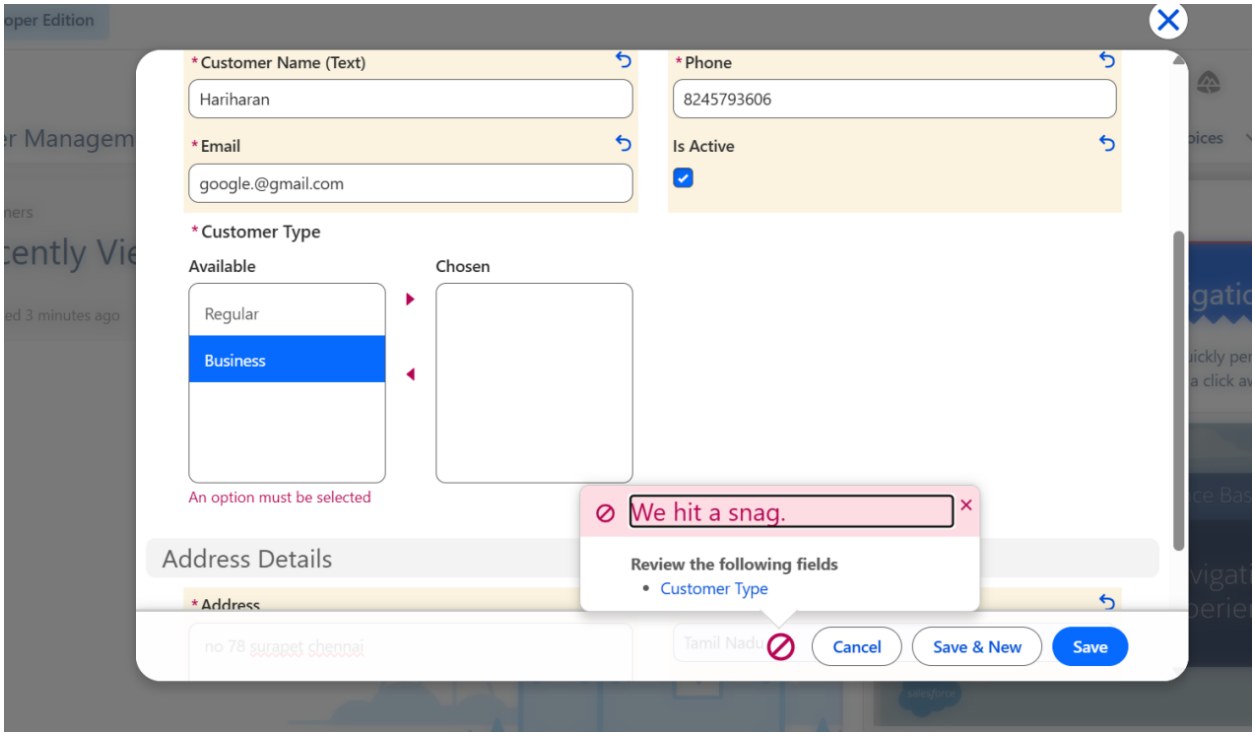


Figure 34 – Validation test evidence.

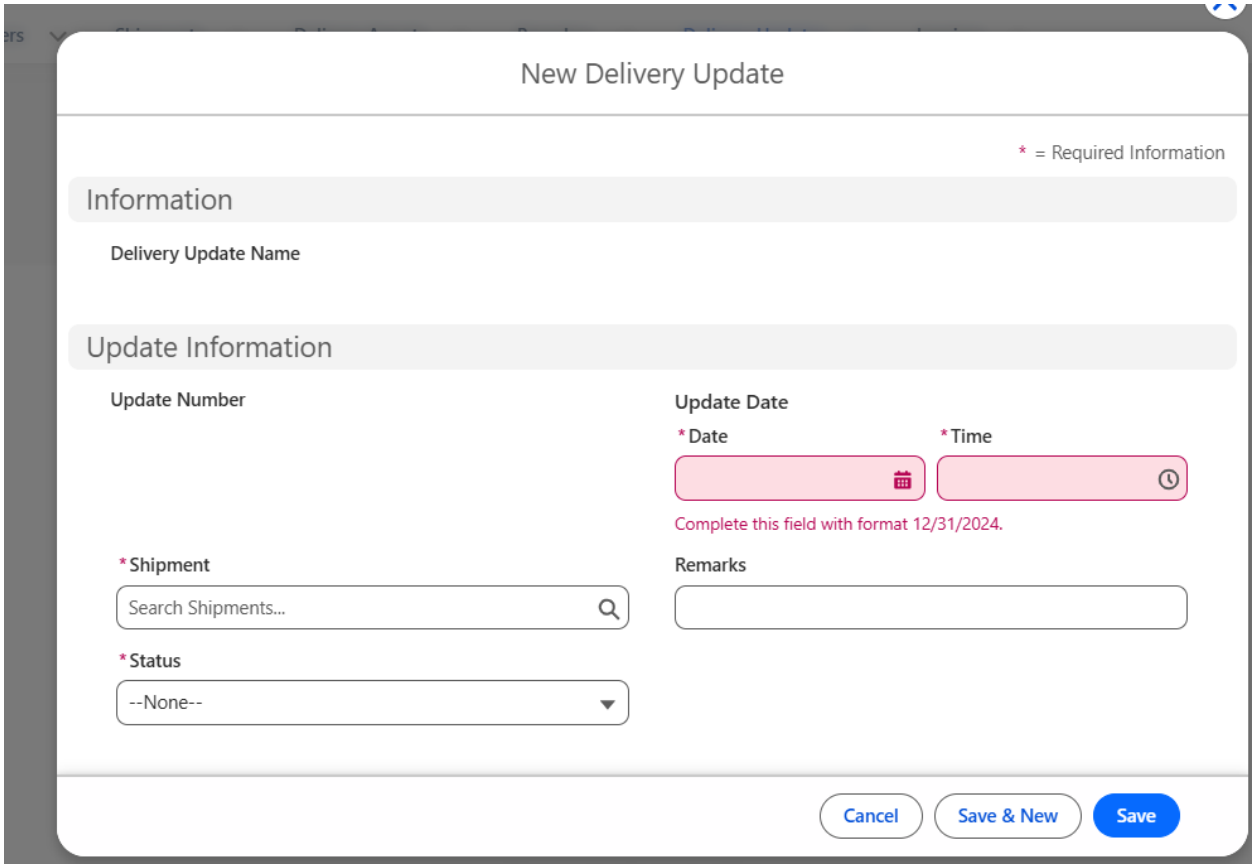


Figure 35 – Delivery Update test/input evidence.

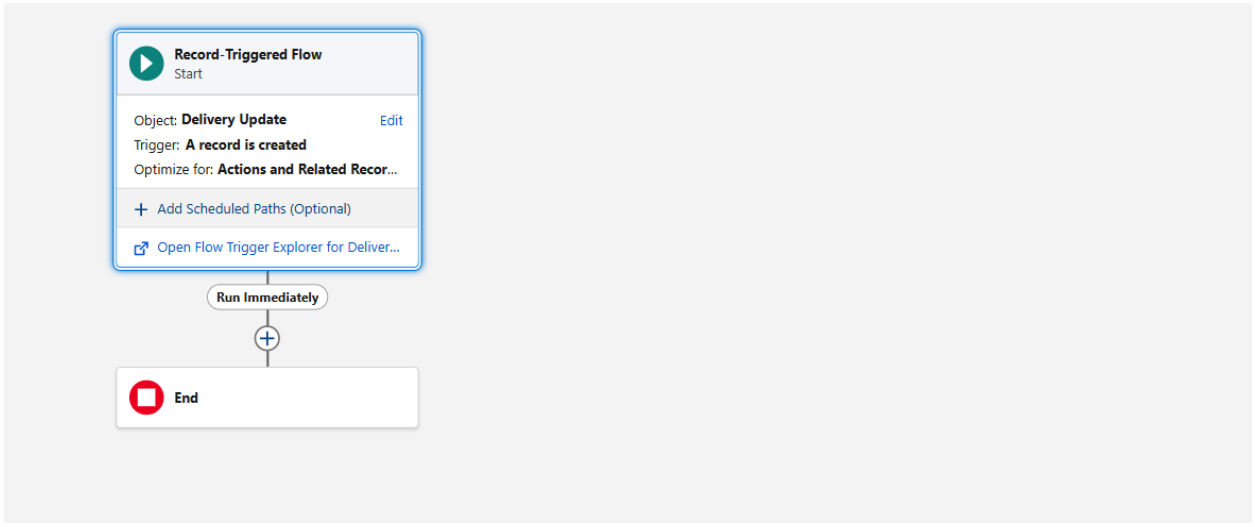


Figure 36 – Flow automation evidence used during testing.

15. CONCLUSION

The Courier Management System was implemented in Salesforce through a sequence of object creation, field configuration, relationships, validation, page-layout design, automation, application configuration, security setup, reporting and dashboard phases. The title-wise evidence in this document places screenshots directly under the implementation topic they demonstrate, making the project completion process traceable from the initial setup through testing and reporting.

Appendix – Screenshot Evidence Index

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